
TUTORIAL ABSTRACT BOOKLET

Asian Modelica and FMI Conference 2026



2026-8-20

CONFERENCE ORGANIZATION COMMITTEE

166 Shuanghongqiao ST, Pingyao, Yuhang District, Hangzhou, China

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SMArtInt+ - Hands-on AI: Generation and Integration of Neural Networks and Surrogate Models into Modelica

Duration: 3 hours

Tim Hanke, XRG Simulation GmbH

Abstract

There is an increasing need for hybrid models consisting of physical and machine-learning or surrogate model parts. XRG has developed the **SMArtInt+** Library, which enables the easy generation and integration of neural networks from different sources and types. This tutorial provides a hands-on introduction to the use of this library, which also includes a Python toolbox. The main use case is the creation and integration of a transient surrogate model for a thermodynamic application.

The tutorial focuses on the generation and utilization of training data, as well as the definition and training of neural network architectures, with the objective of developing high-performance surrogate models. Participants will be introduced to different concepts and approaches for creating data-driven surrogate models and will gain an understanding of the underlying methodologies and best practices.

A central part of the tutorial is the integration of the trained neural networks and surrogate models into Modelica models using the **SMArtInt+ Interface-Blocks**, allowing seamless substitution of physical components with learned surrogates. Additionally, several concepts of hybrid modeling that combine physical and data-driven approaches are introduced, with a strong focus on the practical integration of trained neural networks and surrogate models within the Modelica environment.

Finally, model performance is evaluated through comparisons with the original physical model.

Expected experience of participants:

- Basic knowledge of neural networks/AI and Python required
- Basic Modelica knowledge

Software requirements:

- Windows 10 or 11
- Dymola 2026 Refresh 1 or OpenModelica 1.26 (backup)
- Python IDE (PyCharm preferred)

Link to further information

<https://xrg-simulation.de/en/seiten/smartint>

FMI Beginners' tutorial - Exporting, Simulating, and Co-Simulating FMUs

Duration: 2.5-3 hours

Christian Bertsch, Robert Bosch GmbH; FMI Project Leader; Modelica Association Board Member

Abstract:

The Functional Mock-up Interface (FMI) standard plays a central role in enabling interoperability between simulation tools, particularly in cyber-physical and embedded system development. This tutorial introduces the core concepts of FMI and provides hands-on experience in exporting, simulating, and co-simulating Functional Mock-up Units (FMUs). Designed for both beginners and intermediate users, the tutorial is divided into three main sessions. The first session presents the motivation and history of FMI, key technical concepts, and an overview of FMI 3.0, followed by

live demonstrations of FMU export using Dymola and Simulink. The second session dives into working with FMUs using Python and the FMPy library, including validation, simulation, and integration with Jupyter notebooks. Finally, we will set up co-simulations involving multiple FMUs, highlighting both graphical and command-line workflows, and demonstrating the impact of co-simulation parameters on accuracy and stability.

The tutorial concludes with a joint session summarizing the role of FMI in digital twins and highlighting real-world applications. Throughout the tutorial, attendees will have opportunities to ask questions and follow along with guided examples and shared materials.

Expected experience of participants

Basic familiarity with simulation concepts

Some programming experience (e.g., in Python)

Optional: Experience with Simulink, Modelica Tools, or co-simulation tools is beneficial but not required

Software requirements:

Python with fmpy[complete] ([installation guide](#))

Optional: Matlab/Simulink

Optional: a Modelica tool such as OpenModelica or Dymola to export FMUs.

Optional: A Google account to use Google Colab for co-simulation notebooks

Link to further information: <https://github.com/modelica/fmi-beginners-tutorial-2026/>

OpenModelica Introduction and Hands-on Practice

Duration: 2.5 hours

Cuiting Peng, State Key Laboratory of Nuclear Power Safety Technology and Equipment, State Key Laboratory of Nuclear Power Safety Technology and Equipment, China General Nuclear Power Engineering Co.,Ltd.

Abstract:

This tutorial provides a comprehensive overview of the OpenModelica architecture and development environment. It covers the basics of the Modelica language, OMEdit GUI operations, and model compilation principles. The hands-on session will guide beginners through component library usage and drag-and-drop modeling to complete a full workflow from environment setup to simulation execution.

Software Requirements: OpenModelica 1.26.3

More Information Link: <https://product.dangdang.com/12449354467.html> (The introductory books we published)